



Original paper

An international survey of imaging practices in radiotherapy

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ABSTRACT

Improvements in delivery of radiation dose to target tissues in radiotherapy have increased the need for better image quality and led to a higher frequency of imaging patients. Imaging for treatment planning extends to function and motion assessment and devices are incorporated into medical linear accelerators (linacs) so that regions of tissue can be imaged at time of treatment delivery to ensure dose distributions are delivered as accurately as possible. A survey of imaging in 97 radiotherapy centres in nine countries on six continents has been undertaken with an on-line questionnaire administered through the International Commission on Radiological Protection mentorship programme to provide a snapshot of imaging practices. Responses show that all centres use CT for planning treatments and many utilise additional information from magnetic resonance imaging and positron emission tomography scans. Most centres have kV cone beam CT attached to at least some linacs and use this for the majority of treatment fractions. The imaging options available declined with the human development index (HDI) of the country, and the frequency of imaging during treatment depended more on country than treatment site with countries having lower HDIs imaging less frequently. The country with the lowest HDI had few kV imaging facilities and relied on MV planar imaging intermittently during treatment. Imaging protocols supplied by vendors are used in most centres and under half adapt exposure conditions to individual patients. Recording of patient doses, a knowledge of which is important in optimisation of imaging protocols, was limited primarily to European countries.

1. Introduction

The aim of radiotherapy is to treat a volume of tissue containing a tumour and the surrounding region that may have been infiltrated by

malignant cells. In the ideal treatment, the radiation dose given to the tumour would be sufficient to kill all cancer cells, while that to normal healthy tissue would be low enough to minimise the risk of any serious damage. All the equipment and techniques available are employed in

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trying to maximise the probability of achieving this goal. Most treatments of tumours deep within the body are delivered by medical linear accelerators (linacs) operating at nominal energies between 4 MV and 23 MV and deliver x rays often in 30 or more daily fractions to allow better tolerance of radiation effects by normal tissues.

Treatment beams are now conformed to the shape of the tumour being treated using multi-leaf collimators (MLCs) with individually adjustable motorized high density tungsten alloy leaves. Delivery of shaped beams from multiple directions enables dose distributions to be conformed more effectively to tumours, while sparing surrounding normal tissues. Intensity modulated radiation therapy (IMRT), in which the MLC leaves are moved while x rays are being emitted to create intensity gradients, can further facilitate the creation of complex 3-dimensional dose distributions fitted to the shape of the target volume around the tumour [1,2]. There have been various other developments in treatments performed, most notably Volumetric Modulated Arc Therapy (VMAT), a type of IMRT delivered in one or several arcs [3]. The other ‘paradigm shift’ has been the introduction of stereotactic body radiotherapy (SBRT), in which high doses of radiation are delivered to the target from many angles, requiring high precision that is provided by image guidance and motion management [4–6].

Radiation delivery is most commonly planned by dedicated computers based on three-dimensional (3-D) computed tomography (CT) images of the patient obtained from a CT simulator with patients in positions analogous to those used for treatment and tumour position identified with respect to anatomical land marks or fiducial markers. But improvements in treatment outcomes can only be achieved if the radiation can be delivered to the patient on the treatment couch with the highest level of accuracy. Therefore, imaging is required at the time of treatment delivery to ensure that patients are positioned so that dose distributions will be delivered as planned to the correct tissues. Imaging at the time of treatment also enables plans to be adapted to take account of any anatomical changes that occur. Thus, image guided radiation therapy (IGRT) has become indispensable in the success of radiotherapy (RT) treatment and is now used routinely throughout the world [7–9].

Modern linacs incorporate kV imaging systems that can take planar radiographs or by rotating the x-ray tube and detector around the patient allow reconstruction of cone beam CT (CBCT) scans [10,11]. Imaging can also be performed with the MV therapy sources by using electronic portal imaging devices (EPIDs), especially where kV imaging is not available [12]. More recently magnetic resonance imaging (MRI) modalities and optical imaging systems are becoming available on linac treatment machines in some countries [9,13–15]. Because IGRT allows additional accuracy in localisation of tumour volumes at the time of treatment, the dose delivered to the tumour can be raised, without increasing the dose to surrounding tissues significantly. However, there is a balance to be achieved, since ionising radiation is known to cause cancer [16] and there is evidence that doses delivered from CT scans for medical imaging have an associated risk of cancer [17,18]. Moreover, the imaging performed covers more than the region immediately surrounding the tumour, contributing to risks of secondary cancers in adjacent organs. Thus, improvements in sparing of normal tissue and associated reduction in toxicity through better localisation of tumour targets, must be offset against additional radiation doses to surrounding tissues from imaging, bearing in mind that as the frequency of image guidance is decreased, the margins around the treatment planning volume have to be increased to ensure the whole target is irradiated [19].

It has been assumed in the past that, because doses from imaging were much smaller than therapeutic ones, they were of little concern and as a result less effort was put into their reduction. However, recent changes in patterns of imaging, with more frequent exposures have the potential to give cumulative doses of hundreds of mGy to patient tissues [20–22]. One issue is that unlike the practices in diagnostic radiology, settings for imaging are typically not adjusted let alone optimised for individual patients.

Guidance on the use of IGRT has been published by the American

Association of Physicists in Medicine (AAPM) [23–25], the European Society for Radiotherapy and Oncology [26], and the International Atomic Energy Agency [9], and they are commonly described in clinical protocols. However, surveys undertaken of practices are limited either to single countries [27–29] or specific aspects of treatment imaging [30]. Practices vary between RT centres in the imaging modalities used, including the frequency of imaging, the dose levels used, and the level of optimisation. Some data on this are available for centres in Europe and the United States, but there is less information on how imaging practices are developing in other parts of the world to take advantage of the improvements in treatment delivery. Since this has implications for radiological protection, the International Commission on Radiological Protection (ICRP) has set up a Task Group (TG116) to prepare guidance on the radiological protection aspects of imaging in RT. This survey has been carried out through TG116 to provide information about the use of imaging for RT in different parts of the world.

2. Methods

The ICRP set up a mentorship programme in 2020 to engage university students, early-career professionals and scientists in Task Group activities. Applications were received from a group of medical physicists interested in projects associated with TG116, who came from a broad range of countries in different parts of the world. Since information on the imaging linked to RT in many countries is limited, a decision was made to involve the mentees in a survey of current practices within their countries, to inform the Task Group and assist in the development of guidance. The involvement of any country depended on the nationality or residency of the mentees involved, and because the approach does not follow a random sampling design only descriptive results rather than inferential statistics are presented. However, the survey provided a route for gathering information from a selection of higher-, middle-, and lower-income countries and hence could give a snapshot of practices.

The mentees each established links with RT centres within their countries and encouraged the centres to identify an individual who would complete a survey questionnaire. Additional help in contacting RT centres was obtained from Medical Physics Associations in some countries. The close liaison between the mentees and the RT centres allowed for follow-up and clarification where necessary. The survey took the form of a questionnaire on imaging practices in each centre, formulated by a group of TG116 members. It included 130 separate items and information on the aspects covered is summarised in Table 1.

The survey was conducted on-line using Survey Monkey® and comprised a mixture of numerical questions about RT treatments, imaging equipment, and practices such as the frequency of imaging, and free text fields for comments on reasons for choices, modifications made to protocols, and guidance documents used. The mentees were asked to provide data for a representative selection of centres, but no other attempt was made to target specific centres by the survey organisers. Participation in the survey was organised in two parts. The Mentees were provided with a link that they could send to representatives at each centre with whom they had communicated, through which the representative could register to opt into the survey. This set out the purpose and sought agreement that the representatives could confirm on behalf of their RT centres that they were willing to participate and collected their contact details. Once agreement had been obtained a link to the full survey was emailed to the representative. Since the survey was relatively long, the individual links to the survey were left open, so that representatives could complete sections about practices in their centre over a number of sessions. This also allowed them to consult widely in their departments as questions required multidisciplinary input. The initial links were sent out at the end of July 2020 and the survey closed on 10th November 2020, after which the data were downloaded in a spreadsheet. The number registering an initial interest was 143, but 42 only completed limited sections and so were excluded from the analysis, leaving 101 RT centres that completed the full questionnaire. There

Table 1
Content of survey questionnaire completed by radiotherapy centres.

Pages	Information requested
1	Background
2	Contact information
3	General facility information - Type of facility: Public / Private / Academic - Number of patients treated per month - Percentage of treatments using image guidance - Imaging physicists available for consultation - RT physicists with expertise in imaging - The three anatomical sites treated most frequently
4	Planning X-ray equipment (Nos. of units and vendors) - CT Systems - Conventional X-ray simulator - Other imaging modalities
5	Treatment Imaging Systems - No. of linear accelerators (linacs) and dates of commissioning - No. of Cobalt-60 treatment units - No. of linacs with kV imaging capabilities with make and model
6	Use of on-treatment imaging - Types of imaging procedures used - Use of specific paediatric imaging protocols
7–12	Use of imaging during treatment of specific conditions Frequency of imaging for radical treatments
7	Neuro-oncology / Brain - Head and neck - Breast - Lung - Gynaecology - Prostate
13–16	Protocol use and optimisation for different imaging modalities - Use of protocols as supplied by the manufacturer - Adaptation of protocols for individual patients - Adjustment of jaw/blade settings and fields for paediatric and adult patients - Recording of dose quantities for individual patients - Taking account of imaging dose in treatment plan
13	kV cone beam CT - kV planar imaging - MV cone beam CT - MV planar imaging
17	Additional information - Frequencies for performing QC on imaging systems - Performance tests on: Geometric accuracy, image quality, imaging dose - Use of published guidance for QC protocols - Staff performing QC tests - Working with diagnostic imaging physicist on the optimisation of imaging

were four single data sets from individual countries that were neighbours to the main countries in the survey and these were omitted from the analyses. This left a total of 97 responses from RT centres in nine countries over six continents included in the analysis, and the countries and numbers of complete responses are shown in Table 2. Since the countries participating in the survey represented a wide range in income and healthcare funding, the Human Development Index (HDI) as defined by the United Nations Development Programme [31] was used in

making comparisons and countries are listed in the tables and ordered in the figures in terms of decreasing HDI to bring out differences relating to the level of development. The HDI combines measures of the index of life expectancy, education (literacy rate and receipt of different levels of education), and per capita income. Values move towards 1.0 as the level of development rises. The number of RT centres in the country, taken from the DIRAC database maintained by the International Atomic Energy Agency (IAEA) [32] is also shown in Table 2.

The data were downloaded from Survey Monkey for analysis. The mentees contacted individual RT centres in order to complete data collection where values were missing or clarify apparent anomalies. For example, a few centres indicated that they carried out imaging two or three times per fraction, but on interrogation this was found to relate to images taken at different angles before the start of treatment for some centres, so these were counted as once per fraction.

3. Results

3.1. Radiotherapy centres surveyed

Data obtained from nine countries were included in the analysis. Two RT centres from New Zealand and ten from Australia were combined in an Australasian group as they share a professional organisation, the Australasian College of Physical Scientists and Engineers in Medicine (ACPSEM). Data were obtained from the two radiotherapy centres in Cyprus, reflecting practices over the whole country. In the sample taken, hospitals in the USA, Colombia, and Malaysia were predominantly funded by the private sector, whereas elsewhere public funding predominated or numbers in the two sectors were similar (Table 2). The links with academic institutions varied between regions. The RT centres covered a range of sizes and there were three centres in the USA and one in Australia with ten or more linacs, whereas almost all the centres in Egypt and Malaysia only had one linac (Table 3). Data on the numbers of patients treated at centres within each region are given in Table 3. The median numbers of patients treated per linac per month were between 24 and 50, apart from in Egypt for which the median value was 85. Two centres in Egypt and one in Colombia treated over a hundred patients per month on one linac. The number of patients treated was not associated with HDI.

3.2. Imaging for radiotherapy planning

All the RT centres used CT scans to plan treatments, although there were one or two centres within every region apart from Australasia that also had conventional fluoroscopy/radiography simulators that replicate the geometric beam delivery configuration of a radiotherapy linac. These were widely used before the advent of CT and may have benefits for certain types of treatment. The CT scanners were split between the major vendors, with Siemens having the majority in European countries and Egypt, all scanners in Algeria being from General Electric (GE), the majority in Malaysia being Canon/Toshiba, and half of those in Australasia and Saudi Arabia being from Philips. Those in USA and

Table 2
Data on radiotherapy centres completing the questionnaire for each country, types of centre, numbers of linear accelerators and the numbers of patients treated.

Country	HDI	No. of RTCs in survey	No. of RTCs in country*	Public hospitals (%)	Private hospitals (%)	Academic connection (%)
Germany	0.947	10	293	80	20	50
Australasia	0.944	12	111	67	33	8
USA	0.926	30	2,555	37	63	57
Cyprus	0.887	2	2	50	50	0
Saudi Arabia	0.854	4	14	100	–	75
Malaysia	0.81	7	29	14	71	29
Colombia	0.767	14	58	29	71	14
Algeria	0.748	9	16	67	33	–
Egypt	0.707	9	72	44	44	33

RTC – RT centre, * Data from DIRAC [32].

Table 3

Numbers of patients treated at each RT centre, numbers per linac, and Human Development Index (HDI) for each country or region.

Country	Linacs in RTCs in survey	Median no. of linacs per RTC	Median patients / month at centre	Range patients / month at centre	Median patients / month on linac	Range patients / month on linac
Germany	26	2.5	150	60–300	50	30–100
Australasia	55	3	97	30–550	29	17–73
USA	143	4	116	20–140	23	12–75
Cyprus	5	2.5	95	80–110	38	36–40
Saudi Arabia	16	2.5	105	65–200	45	13–67
Malaysia	9	1	35	9–85	25	9–85
Colombia	24	2	70	30–250	40	20–120
Algeria	24	3	55	20–180	24	7–80
Egypt	17	1	180	50–400	87	50–250

Colombia were split between vendors. The choice of vendor to some extent reflects geographical and cultural links.

Other imaging modalities used in the planning process are shown in Fig. 1, with about 80%–90% of centres in the Germany, Australasia, USA, and Cyprus using MRI and 65%–80% using positron emission tomography (PET)/CT, and 40%–50% in the Asian and South American countries. There was a general trend for the proportion of centres using these options declining with the HDI of the country. There was limited use of single photon emission CT (SPECT) or ultrasound. Some of the imaging modalities used were sited elsewhere in the hospital complex, but between 20% and 50% of hospitals in the European, African, and American countries had MRI scanners within the RT centres, although only one Australian RT centre had an MRI scanner in the department and there were none in Malaysia. Between 10% and 30% of RT centres in the European and American countries had PET/CT scanners within the centre, but only 10% in Australasia and African countries and again none in Malaysia.

3.3. Radiotherapy treatment imaging

The median numbers of linacs per RT centre were four in the USA, three in Australasia and Algeria, 2.5 in the European countries and Saudi Arabia, two in Colombia, and one in Malaysia and Egypt (Table 3). However, several countries had individual centres with larger numbers, with one centre in Australasia having 16 linacs, one in Saudi Arabia having nine, and three centres in the USA having 11, 12, and 13 linacs respectively. The linac models were split between two vendors with Elekta predominating in European and African countries and Varian in the American and Australasian ones. The majority in Europe and Africa had been commissioned during the last four years, with 10% or less being over ten years old, whereas in Australasia, Colombia, Malaysia, Saudi Arabia, and the USA only 30%–40% had been commissioned during the last four years, about 30% were over ten years old, and some were more than 15 years old (not in Australasia). kV imaging facilities

were available on 80%–96% of linacs in all countries, apart from Colombia (71%) where most centres had one linac with kV imaging and Egypt (15%) where only two linacs out of 17 had kV imaging.

All RT centres used IGRT for the majority of patients, and in eight countries most RT centres used it for over 75% of patients treated (Fig. 2). A smaller proportion of RT centres in Egypt used IGRT most of the time, but most centres only had MV imaging facilities. Each RT centre was asked to list the three treatments performed most frequently with IGRT and these are shown in Fig. 3, with the breast and prostate being predominant, especially in countries with higher HDIs, followed by head and neck.

There is a range of imaging options available for IGRT to confirm registration of the target immediately prior to treatment and the availability of several different imaging modalities has improved management of geometric uncertainties in radiotherapy practice. Orthogonal pairs of planar images can be recorded using the MV treatment beam with EPIDs, with kV imaging systems integrated into linacs, or combined in kV/MV pairs for fast acquisition, and enable assessment of patient positioning in all three cardinal directions. Single direction kV or MV imaging allows assessment of motion using fluoroscopy or Cine MV imaging. CBCT images may be recorded using the kV or MV imaging systems, although the latter can involve significantly higher doses to patients, or a separate CT scanner slid into position on rails can be used. Stereotactic imaging systems include additional translational and rotational parameters for patient positioning. Fig. 4 shows the range of imaging options that are used in RT centres surveyed. CT based systems have the advantage of being directly compatible with the planning CT and as such kV CBCT is the option used most frequently by the majority of RT centres, but orthogonal image pairs are also used regularly. For those RT centres using kV planar imaging 20% – 30% indicated that they also used live fluoroscopy, while 70% used fluoroscopy in Germany. The only country in which kV CBCT was employed in less than 80% of centres was Egypt, in which the use of MV-MV pairs was predominant because of the lack of kV imaging equipment. All four RT centres in Saudi Arabia indicated that they used the MV-kV pair technique that can save time in treatment delivery, as it can be performed without gantry rotation.

Non-ionising radiation techniques were used to make checks during treatment with more than half the centres in Germany and the USA using optical surface guidance, Australasia and Saudi Arabia centres using ultrasound predominantly, and a few centres using MRI, where this was available (Fig. 5). Countries at lower HDI levels tended to have fewer options available.

It was apparent that kV CBCT was the imaging modality used most frequently in all countries surveyed apart from Egypt (Fig. 4). The imaging equipment operated and the availability of experienced staff to perform the more specialised IGRT procedures depends strongly on funding reflected in the HDI [31] (Table 2). It is apparent from Fig. 6 that the proportions of RT centres with CBCT imaging capabilities relevant to IGRT falls off with HDI, and higher HDI countries use less kV planar imaging. All RT centres in countries with HDIs of 0.8 and above

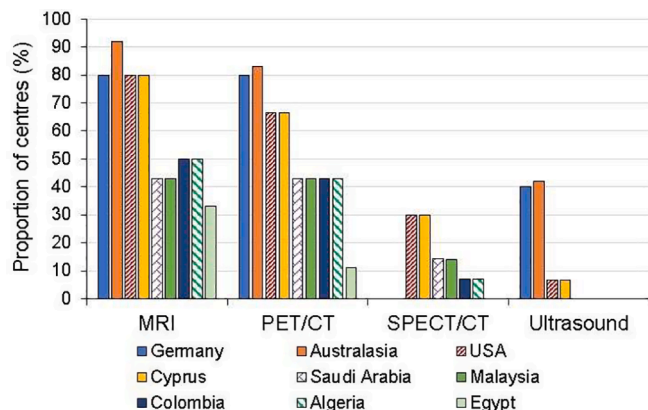


Fig. 1. Imaging modalities used in the planning process in addition to CT.

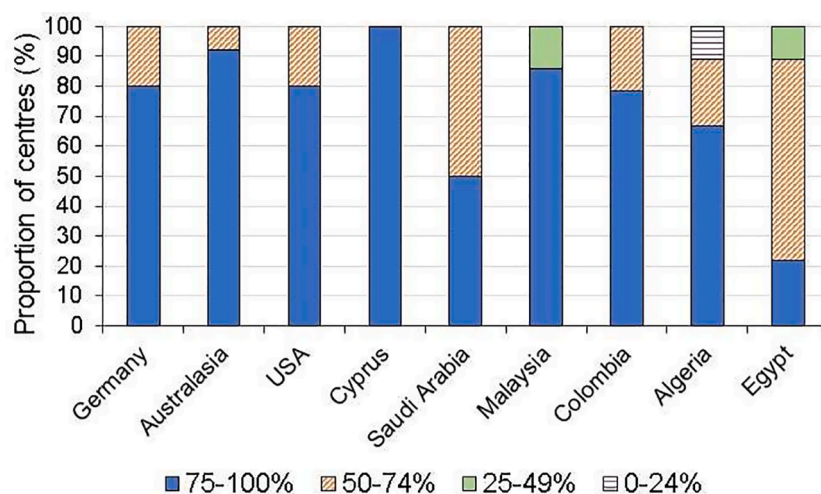


Fig. 2. Proportions of treatments for which image guidance is used in RT centres, in terms of the percentage of RT centres in each country within each range.

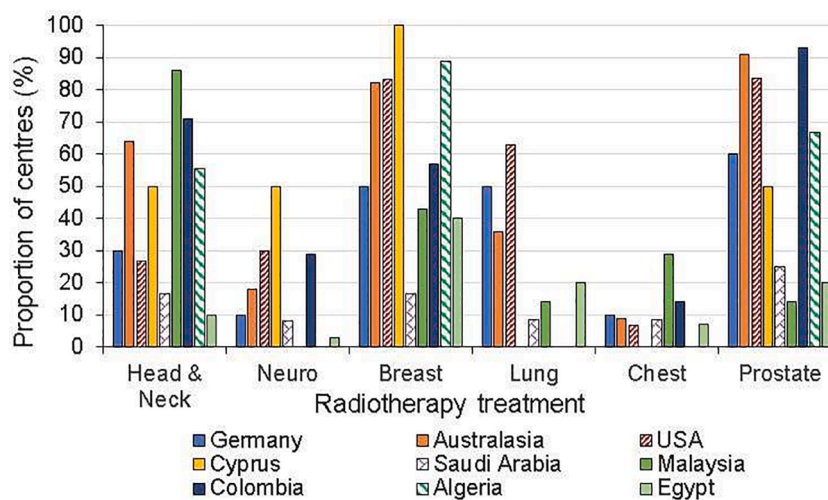


Fig. 3. The three treatments performed most frequently by RT centres within each country in terms of the percentage of RT centres.

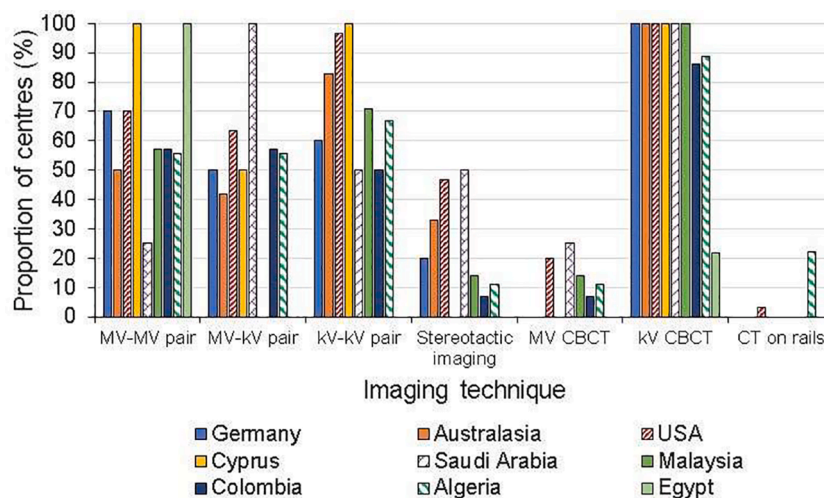


Fig. 4. Use of image guidance options by RT centres in the different countries.

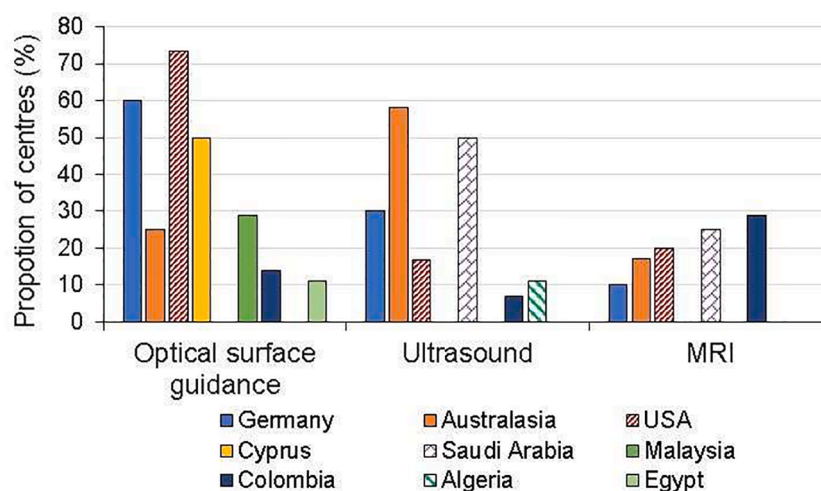


Fig. 5. Non-ionising radiation options used during delivery of RT treatments in different countries.

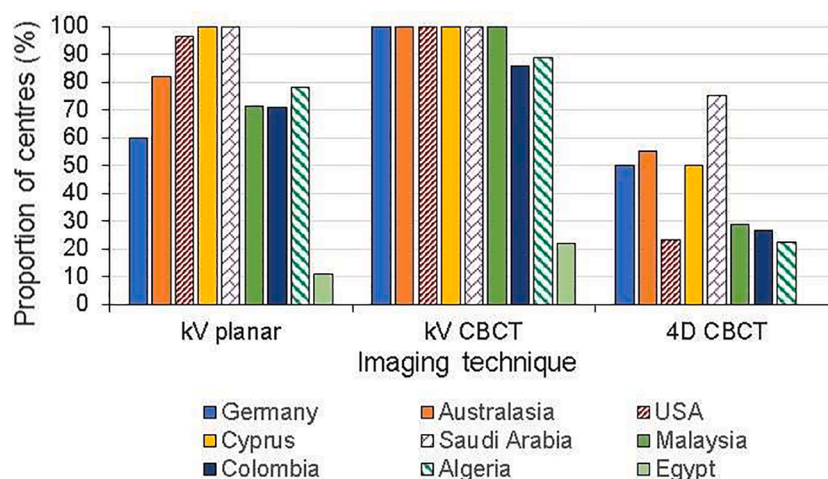


Fig. 6. Proportions of RT centres within countries that have available and make use of different kV imaging capabilities.

have kV imaging equipment linked to at least some linacs, but the proportion declines for countries with lower values of HDI. However, although the majority of centres used CBCT, only in Germany, Australasia, Cyprus, and Saudi Arabia did 50% or more use 4D CBCT for respiratory controlled treatment delivery, while in the remainder of countries, including the USA, this was only used by 20%–30% of RT centres (Fig. 6). Although this is a very limited survey, it nevertheless shows a link between funding and the availability of equipment and techniques for IGRT.

3.4. Frequency of imaging during treatment

In the questionnaire separate pages were included for six treatment regions (Table 1). The RT centres were asked to list all the imaging options that they would employ for radical or curative treatment of each and the ones used most frequently. Proportions of RT centre using various options for the frequency of imaging for neurological and other treatments of the head are given in Table 4, and for treatments of the body in Table 5. Although practices varied with the location of the tumour being treated, general trends in practices within individual countries tended to follow similar patterns for most types of treatment. A pattern that is immediately apparent is that the frequency of imaging declines as one moves down both tables to countries with lower HDIs.

90–100% of the RT centres in Australasia and the USA perform imaging at every fraction, while in Cyprus most use either once per fraction

or three times per week. In Germany the approach of imaging once per fraction was adopted more frequently, but a range of other options was used, while in Saudi Arabia imaging at each fraction was used by more centres, but imaging for the first three fractions, followed by weekly imaging was also employed. A range of imaging practices from once per fraction to once per week was used by centres in Malaysia, Colombia, and Algeria, with the option of the first three fractions, followed by weekly imaging being used by many. In countries using a range of alternative approaches, VMAT and IMRT treatments were generally imaged at each fraction. Some centres in Germany, Australasia, and the USA performed two sets of imaging during SBRT treatments of the brain (Table 4), as did some centres in Saudi Arabia and Australia during lengthy SBRT lung treatments (Table 5), with the second image set being taken half way through treatment to verify position. One US centre used dual imaging in each fraction that consisted of acquiring both CBCT and planar image pairs and this was recorded as imaging twice per fraction.

The treatment imaging options used more often in Egypt were once per week or once per course of treatment (tables 4 and 5). The less frequent use of imaging in Egypt is likely to be associated with the need to rely on MV imaging because of the limited availability of kV facilities (Fig. 4). Fig. 7a shows the proportion of centres where patients are imaged at least at each fraction for treatments of four regions of the body. This is between 30% and 100% for most countries, but only 10% or less in the Egyptian RT centres. Fig. 7b gives the median number of patients treated per linac per month and shows that Egypt carries out

Table 4

Proportions of RT centres surveyed in each country using different options for frequency of imaging for radical treatments of the head given as a percentage of centres surveyed.

Country	Treatment	Twice per fraction	Once per fraction	3 time per week	First 3 fractions then weekly	Once per week	Once per course
Germany	Brain	10	60			30	
	Head & neck		60	10		30	
Australasia	Brain	27	73				
	Head & neck		91			9	
USA	Brain	9	91				
	Head & neck	3	97				
Cyprus	Brain*		50				
	Head & neck		50	50			
Saudi Arabia	Brain		75		25		
	Head & neck		100				
Malaysia	Brain		29	14	43	14	
	Head & neck		29	14	43	14	
Columbia	Brain			14	36	50	
	Head & neck		36	14	29	21	
Algeria	Brain		44	22	33		
	Head & neck*		78			11	
Egypt	Brain			14	36	50	
	Head & neck			11	11	66	11

*The option “other” was given for the treatment option used most frequently.

Table 5

Proportions of RT centres surveyed in each country using different options for frequency of imaging for radical treatments of the trunk given as a percentage of centres surveyed.

Country	Treatment	Twice per fraction	Once per fraction	3 time per week	First 3 fractions then weekly	Once per week	Once per course
Germany	Breast		50		20	30	
	Lung		50	20	10	20	
	Gynaecology		50		20	30	
	Prostate		70	20	10		
Australasia	Breast		91		9		
	Lung	9	91				
	Gynaecology		100				
	Prostate		100				
USA	Breast	3	97				
	Lung*	3	93				
	Gynaecology	3	90	6			
	Prostate	3	97				
Cyprus	Breast		50	50			
	Lung		50	50			
	Gynaecology		50	50			
	Prostate		50	50			
Saudi Arabia	Breast		50		50		
	Lung	50	50				
	Gynaecology		75		25		
	Prostate		75		25		
Malaysia	Breast		43		43	14	
	Lung		57		29	14	
	Gynaecology		43		43	14	
	Prostate		71		14	14	
Columbia	Breast		14	14	14	57	
	Lung		29	29	21	21	
	Gynaecology		21	29	29	21	
	Prostate		36	21	21	21	
Algeria *	Breast		55		11	22	
	Lung		78			11	
	Gynaecology		67			22	
	Prostate		89				
Egypt	Breast				11	22	66
	Lung				11	77	11
	Gynaecology				22	11	66
	Prostate			22	11	55	11

*One centre responded “other” for options used most frequently for all treatments.

less imaging, but treats more patients per linac. Australasia, USA, Malaysia, and Algeria, which tend to carry out more imaging than other countries, are the ones that treat fewer patients per month on each linac.

3.5. Optimisation of imaging protocols

Preliminary consultation had indicated that few centres were able to

provide data on doses from imaging, so the questionnaire sought information on the use and adaptation of protocols. The adoption of paediatric protocols in RT centres that carried out paediatric procedures was variable (Fig. 8), for kV CBCT being 100% in Cyprus, about 90% in Colombian RT centres, 70% in Australasia, Saudi Arabia, and Malaysia, 40% in USA, and 30% in Algeria and Germany. Paediatric protocols were used in all centres for kV planar imaging in Malaysia and Saudi

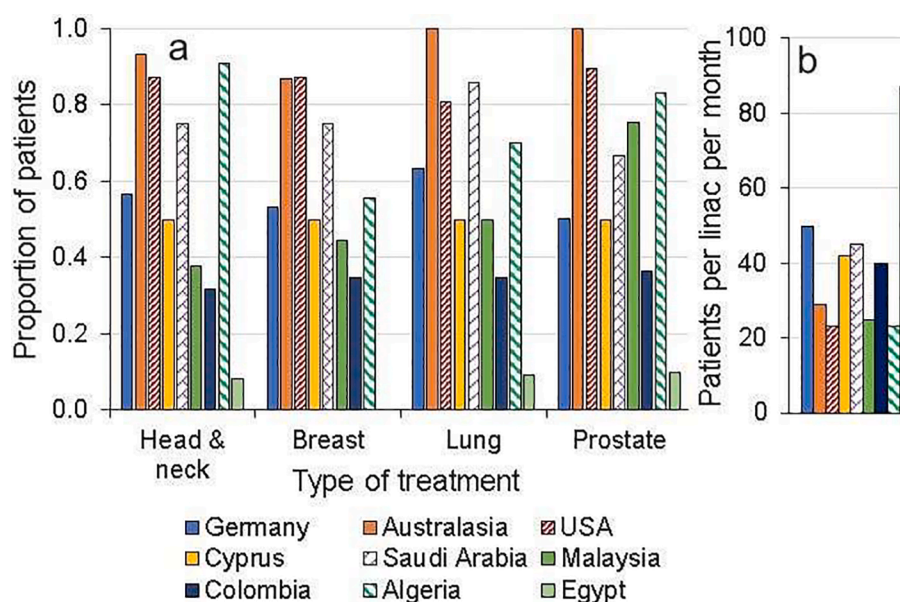


Fig. 7. a) Proportion of patients imaged at least once per fraction for four treatments by country and b) the median value for the number of patients treated per month per linac in data sets from each country.

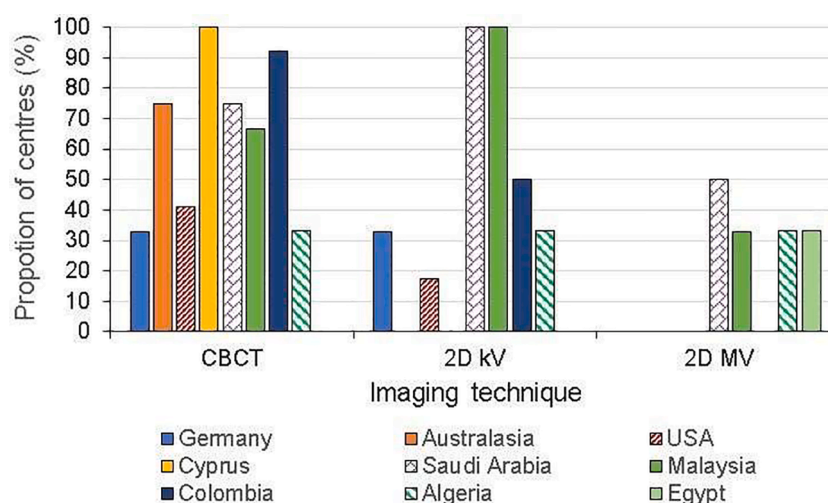


Fig. 8. The application of paediatric protocols for different IGRT imaging in centres carrying out treatments of children.

Arabia and by 50% of centres in Colombia. The reason why paediatric protocols were not used, especially by some of the countries with higher HDIs is uncertain, but may reflect adaption of techniques to individual patients rather than reliance on protocols supplied by the vendor.

The questionnaire sought information on the extent to which kV CBCT imaging protocols supplied by the vendor were simply employed unaltered or whether some optimisation was undertaken by the RT centres, and results are shown in Fig. 9. 90–100% of RT centres in Algeria, Cyprus, Colombia, Egypt, Malaysia, and the USA used the adult vendor protocols as the basis for imaging, although 10–40% modified them for some individuals. While in Australasia and Germany, the vendor protocols were only used by just over 50% of RT centres. The types of modification reported were adjusting the mA and kV to optimise performance, sometimes based on imaging of standard phantoms, increasing the kV and/or mAs for large patients, adjusting the field size, and changing the number of acquisition frames in chest scans. The majority of RT centres, which carried out paediatric treatments in all countries, adjusted the kV CBCT jaw blades for individual children. The proportions making adjustments in jaw blade settings for adults were

generally slightly lower than for children. The patterns in use of vendor protocols were similar for kV planar imaging with 30%–60% of centres making adjustments to the field size and MV planar imaging, with 30%–90% of RT centres adjusting the field size.

Fig. 10 shows that doses from kV CBCT and planar imaging were recorded by 50% of responding RT centres in Cyprus, Germany, and Saudi Arabia. This was predominantly in terms of the CT dose index (CTDI) for CBCT and either the exposure factors or the kerma-area product (KAP) for planar imaging. Elsewhere the recording of dose information was limited. The recording of patient doses from imaging requires up to date equipment, funding, and experience, but this practice did not link to HDI level. The Australasian centres in the survey did not record doses for kV CBCT or kV planar imaging and in the USA only one centre recorded kV CBCT doses.

The recording of patient dose is an important early step in the implementation of optimisation, and another is the involvement of medical physicists with expertise in diagnostic imaging in the optimisation process or improving the knowledge of RT physicists in imaging dosimetry and optimisation. The proportion of centres in the different

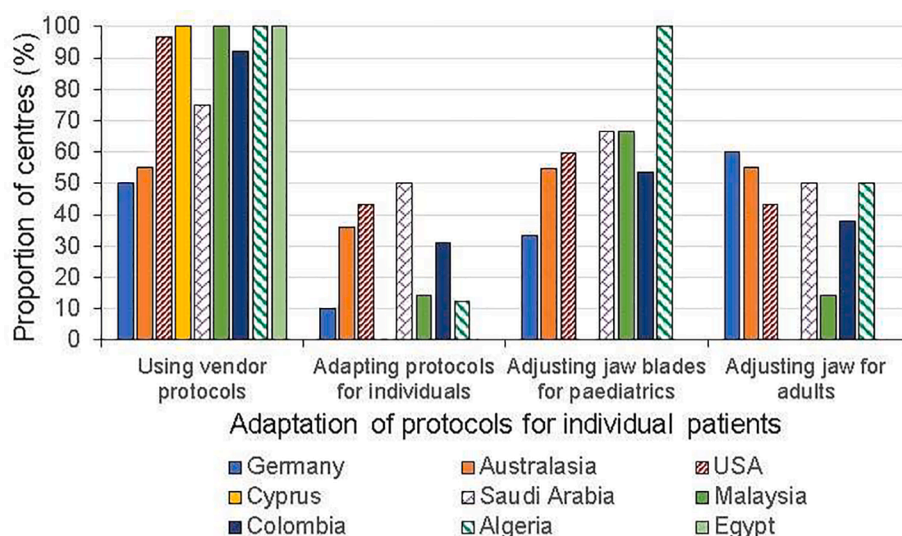


Fig. 9. Histograms showing the use of vendor protocols for kV CBCT and adaptations made for individual patients. The data for paediatric procedures only relate to centres that treat children.

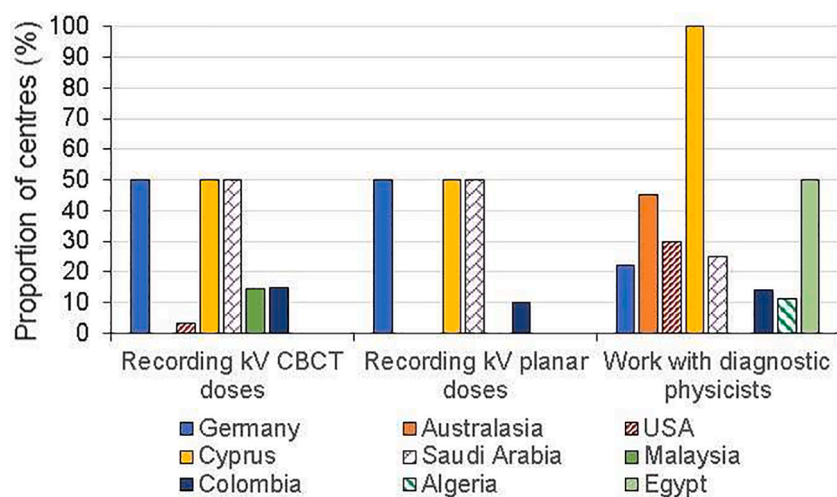


Fig. 10. Proportions of centres recording patient doses for kV imaging and involvement of diagnostic physicists in optimisation of imaging for IGRT in RT centres.

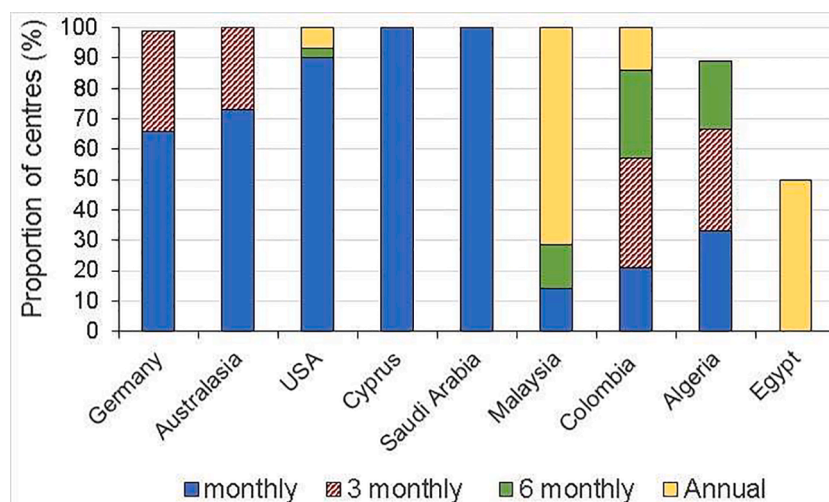


Fig. 11. Frequency of QC testing of kV imaging devices on linacs in different countries.

countries that used diagnostic physicists was variable (Fig. 10) and the involvement of physicists with diagnostic experience, appears to lag behind the acquisition of kV imaging equipment (Fig. 6).

3.6. Quality control of imaging equipment

The last section of the questionnaire sought information on performance of QC on imaging equipment and the healthcare professionals involved. The frequency with which QC was carried out on kV imaging equipment varied from monthly to annually (Fig. 11). The frequency tended to decline with the HDI of the country and QC tests were not performed at all on imaging equipment in some centres in the two African countries. All centres with kV CBCT performed QC on geometric accuracy, and the majority also checked image quality, apart from Colombia where this was only checked by 30% of RT centres. 100% of RT centres in Cyprus, Germany, and Saudi Arabia, and 85% in Malaysia measured the dose performance, but this was only done by 30–60% of the remaining countries. The majority of centres used published guidance, most frequently reports of the AAPM primarily Task Groups 142 and 179 [15,24]. Other guidance documents referred to were DIN standards (Germany), requirements of the Ministry of Health (Malaysia), guidance provided by vendors, French Guidance [33] (Algeria), guidance from the American College of Radiology, and instructions linked to performance evaluation tools. Tests were performed by different healthcare staff groups with medical physicists making up the largest proportion (Fig. 12), some being trained diagnostic physicists and others therapy physicists adapting their skills. Some RT centres also reported that tests were carried out by third party companies or equipment engineers, and not by medical physicists or radiographers within the RT centre. The amount of performance testing appears to be generally increasing with a number of RT centres commenting on changes they were introducing currently or planning for the near future.

4. Discussion

4.1. Survey coverage and treatment workloads

This paper reports results from an international survey of imaging practices in RT across nine countries on the radiological protection aspects of imaging in RT. The number of published surveys gathering information on imaging practice in different countries is limited. Anastasi et al [30] reported a survey on imaging relating to motion management, from hospitals around the globe, but this involved primarily data from high-income countries. The present survey, although not collecting any data on patient doses, provides information about the availability of imaging equipment and practices in its use in countries from different parts of the world with a range of incomes and stages of service

development. The survey allowed the ICRP mentees to find out about imaging practices for RT in their own countries and how they compared with those employed elsewhere, but since it only includes a limited number of sites in a few countries, it only gives a snapshot of practices.

The number of patient treatments carried out each month varied widely with Egypt having the highest median values of patients per linac (Table 3). The high numbers are likely to be associated with a number of factors, including the less frequent performance of imaging (Fig. 7), but also the adoption of extended working hours where there is limited availability of linac facilities. For instance, 120 patients are treated each month at one hospital in Colombia by employing a series of shifts that extend operation times from 6 am until 12 midnight each day.

4.2. Use of imaging

All RT centres involved in the survey used CT scans to provide the images required for treatment planning and most centres in the four countries with the highest HDI levels used results from MRI and PET/CT in the planning process (Fig. 1). Image guidance is used for between 75% and 100% of treatments by the majority of RT centres in eight out of nine countries in the survey (Fig. 2), with 80%–100% of RT centres using kV CBCT (Fig. 4), confirming that this has become the dominant image guidance modality. This is likely to be due to the volumetric nature of the imaging and the similarity in format to the CT scans used for planning. However, CBCT is one of the higher dose imaging modalities and, with the increasing use of daily imaging for treatments that may extend to more than 30 fractions, consideration of dose implications and the need for optimisation becomes ever more important.

Most RT centres in Egypt made less use of image guidance (50%–74% of treatments) (Fig. 2), but only two out of nine centres had kV CBCT, and here the predominant imaging technique used was MV planar imaging (Fig. 4). This poorer provision of imaging capabilities is likely to be linked to the available funding, with Egypt having the lowest HDI (Table 2). However, less frequent use of imaging may allow larger numbers of patients to be treated (Fig. 7b) and so probably represents the most effective use of available resources, with the additional time required for imaging with the MV facilities available, for which image quality is poorer and radiation dose higher, being hard to justify. Various non-ionising radiation options were employed to aid IGRT of which optical surface guidance and ultrasound were the more frequent (Fig. 5). The impact of this technology at this point in time is difficult to assess but it is a rapidly growing IGRT option [34,35].

The majority of RT centres use kV imaging, but the proportions applying more specialised techniques declined with the HDI level of the country (Fig. 6). For example, 4D CBCT was only used by 20%–30% of RT centres in countries with lower HDIs. Motion management during planning and treatment has significantly improved RT practice and

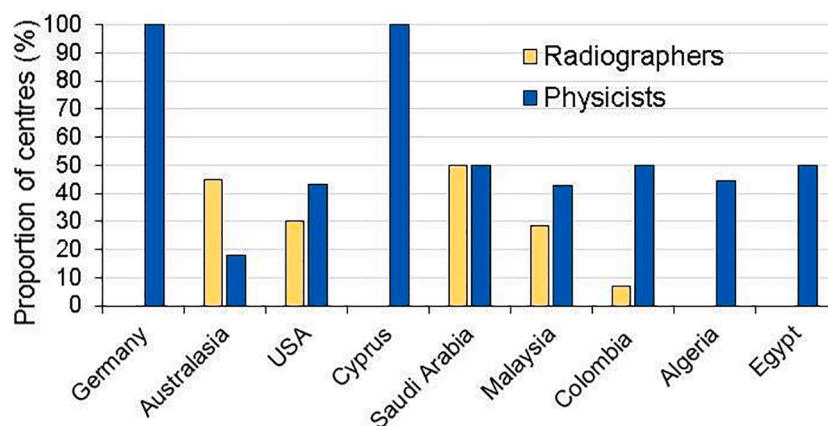


Fig. 12. Healthcare staff carrying out performance tests on imaging equipment.

made techniques such as stereotactic body RT possible, which was used by 10%-40% of RT centres in several countries (Fig. 4). Techniques of this type that rely on oversampling in CT will necessarily deliver higher doses to patients than standard imaging. A recent survey of imaging practice has shown that significant numbers of RT centres in higher income countries use intra-fractional real time respiratory motion management [30], which is recommended by the AAPM when respiratory motion exceeds 5 mm [36]. The introduction of these techniques on a wider scale should be accompanied by the development of standard approaches, recommendations, and guidance.

The frequency with which imaging was used during treatment varied to some extent with the tissue being treated (Tables 4 and 5), but depended more on practice in the RT centre or country. Overall the frequency of imaging was lower in countries at a lower level of HDI. 90%-100% of RT centres in Australasia and the USA used imaging at every fraction for almost all treatments, 50% or more of RT centres in Germany, Cyprus, Saudi Arabia and Algeria use imaging at every fraction, but proportions vary with type of treatment in Malaysia and Colombia. On the other hand, in Egypt imaging once per week or once per course of treatment were the options used more frequently. Imaging at each fraction may be justified when it allows for margin reduction [19,37] and as such higher doses to be delivered with precision while sparing normal tissue. Improving precision through imaging becomes particularly important in techniques where treatment is delivered in fewer higher dose fractions. The survey did not probe the treatment intent specifically, apart from the frequencies of imaging linked to radical treatments of particular regions of the body. However, IGRT tends to be employed more frequently for treatments with curative intent. Differences in patient presentation, such as a larger percentage of advanced cancer stages in low- and middle-income countries, would have an effect on utilisation of imaging.

4.3. Optimisation of imaging protocols

Vendors provide protocols with recommended settings and factors for kV CBCT imaging carried out during treatment, and these were used by 90%-100% of RT centres in most countries, especially those with lower HDI levels (Fig. 9). Experience from surveys has shown that vendor protocols may not always be optimised [27,29], and only 50%-60% of RT centres in Germany and Australasia used them, as centres had developed their own optimised protocols. Most x-ray imaging tools in the treatment room only have limited automatic dose optimisation so that many adjustments to shutter size and mAs need to be made manually. 30%-60% of those using vendor protocols reported some level of optimisation, especially for paediatric patients. The fact that many RT centres use the standard protocols supplied places a responsibility on vendors to engage in the optimisation of protocols.

A knowledge of doses that patients receive from imaging is an important prerequisite for carrying out optimisation, because digital images provide little feedback on dose levels. However, doses from imaging have been a lower priority in RT generally. The dose distributions from diagnostic radiology and the quantities used to measure them differ from those used in RT, so have less meaning for RT staff. 50% of RT centres in the European countries and Saudi Arabia recorded imaging doses, but in other parts of the world the proportions were 15% or less (Fig. 10). In diagnostic radiology information on patient doses is considered to be important in the development of programmes for optimisation of radiation protection [38].

Optimisation of imaging practices requires a different area of expertise from RT treatment, and RT physicists in some RT centres commented on the value of working with physicists trained in diagnostic radiology imaging physics. The involvement of diagnostic physicists in optimisation was variable, and was not correlated with the recording of doses, which appeared to be associated more closely with geographical region, linked to the stronger tradition of recording patient doses for diagnostic imaging in Europe (Fig. 10). In many countries diagnostic

physicists may be involved primarily in carrying out QC tests required by regulations, and do not undertake patient dose surveys, or contribute to the selection of exposure conditions and optimisation.

QC tests were performed on imaging equipment by almost all RT centres in the survey, apart from a few of the African centres (Fig. 11), with most carrying out checks monthly on geometric accuracy and image quality. 80%-100% of RT centres indicated that they verified dose indices on the equipment in Germany, Cyprus, Saudi Arabia and Malaysia, but only 40%-60% of the RT centres in other countries reported any measurement of dose performance. The QC tests are performed by a combination of physicists, radiographers, and other groups such as RT engineers not included in Fig. 12. A knowledge and understanding of the results from QC dosimetry measurements is potentially valuable for the optimisation of imaging protocols. This link will not be established if dosimetry measurements are not carried out or may be lost if different groups of staff carry out QC tests and optimisation.

4.4. Dose and dose distribution from imaging

There are fundamental differences in the magnitudes and distributions of radiation doses from diagnostic x rays and high energy therapeutic photon beams from linacs. In the diagnostic energy range, radiation levels are commonly measured in terms of air kerma. The quantity that is measured for radiography is the KAP, which is a product of the incident air kerma and the area of the x-ray beam that provides a measure of radiation entering the body. CT scans irradiate the body from multiple directions, and the dose quantities measured represent values for absorbed doses in air within standard phantoms. The CTDI provides a measurement related to the average dose within head or body phantoms of specific reference sizes, again with respect to air [39]. Values for the CTDIs are multiplied by the length of a CT scan in order to provide a measure related to the total dose from the scan, which is called the dose length product. Monte Carlo simulations can be used to compute absorbed doses from imaging to individual organs and tissues, and these are the only values that can provide a meaningful comparison with absorbed doses to organs adjacent to the treatment volume from radiation therapy. This is particularly relevant for bone where doses from kV radiation are significantly higher due to the larger contribution of the photoelectric effect.

The doses that patients receive from imaging although being much less than the therapeutic dose, make a contribution that is not negligible and typically extends beyond the actual treatment volume, so as treatment methodologies become more refined it is reasonable that these should be taken into account in treatment prescription, although this is not established in treatment planning systems at the present time [28]. It is not easy to obtain doses from kV imaging in a suitable form for inclusion, but doses from MV images are in MU and can in principle be added into the treatment plan. Six RT centres in the survey, out of the 66 that stated they used MV planar imaging, indicated that they were recording and accounting for the doses in treatment plans, three in Germany, two in Algeria, and one in Australia.

5. Limitations of the study

The present data provide information on imaging practices in RT from a limited number of centres in nine countries. The selection of countries depended on the nationality or residency of applicants for positions as ICRP mentees assisting Task Group 116. The overall percentage coverage in the larger, higher income countries was small, being 3% in Germany and 1% in the USA. However, 11–13% of RT centres completed the questionnaire in Egypt and Australia, 24–28% in Colombia, Malaysia, and Saudi Arabia, 56% in Algeria, and 100% in Cyprus. As a result, the survey does provide an indication of practices in low- and middle-income countries for which less information is currently available.

Whether or not the selection of RT centres within each country is

representative of practices across the whole country relies upon the diligence of the local mentees in contacting different departments, and the willingness of individuals within those departments to complete the survey. It cannot therefore be taken as truly representative of practices within a country, as no attempt was made to avoid possible bias. As a convenience sampling approach was used rather than a cluster or stratified random sampling design, descriptive results rather than inferential statistics are presented throughout this report.

The data accuracy relies on self-reporting of practices by representatives at each centre, and although points that appeared anomalous were followed up by the mentees within each country, and cross checks were made of self-consistency of data within the spreadsheets, the accuracy relies on the knowledge and diligence of the representatives completing the questionnaire. Nevertheless, the survey provides a snapshot in time on the application of IGRT in a wide range of countries at various stages in the development and improvement of RT services.

6. Conclusions

A survey of the use of imaging in RT has been carried out via an online questionnaire, involving 97 RT centres from nine countries on six continents. This was done through the ICRP mentorship programme that appointed mentees from each country who liaised with RT centres. The survey aimed to provide a descriptive snapshot of practices around the world rather than a comprehensive survey, in order to inform ICRP Task Group 116 about potential needs that should be addressed in preparation of guidance on the radiological protection aspects of imaging in RT.

Responses showed that all centres used CT scans for planning treatments often supplemented with data from MRI and PET/CT scans. All centres used IGRT and the majority had kV CBCT systems attached to linacs. Trends in availability of equipment were linked to the HDI level reflecting income and level of development, and in Egypt, which had the lowest HDI, only two from nine centres had kV treatment imaging facilities. The majority of RT centres used imaging at every fraction and although there was variation with the location of the tumour being treated, similar practices tended to be followed for most types of treatment within one country. The proportion of centres in countries with lower HDI levels using daily imaging was lower. The less frequent use of imaging in Egypt was associated with a larger treatment workload and a heavier reliance on MV planar imaging, and is likely to represent a more effective use of limited resources. The majority of RT centres used imaging protocols supplied by vendors with under half adapting exposure factors or field sizes to individual patients and optimisation of imaging practices appears to be lagging behind the acquisition of kV imaging equipment. Recording of patient dose information for imaging was limited and in half of the countries only 40%–60% of RT centres included assessment of dose performance in QC tests, but without this information optimisation of imaging protocols will be difficult to achieve. The survey has provided an appreciation of the wide range in imaging practices in countries in different parts of the world. The differences seen between countries and the link to development makes preparation of guidelines and recommendations, such as the planned ICRP TG116 report, important as tools for standardisation.

This paper represents the views of the authors, not necessarily the views of ICRP

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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